

GO SEE

Architecture & Requirement Analysis Document

New process flow: scheduling, supplier threshold, attendance, closure and recirculation

Document purpose

This version completely replaces the previous scheduling and matching workflow with the newly agreed process. It retains the approved engineer registration, supplier registration, category / sub-category, security, and general platform requirements, while adding supplier-response thresholds, procurement override, attendance verification, closure deadlines, recirculation eligibility, reporting, and configurable reminders.

Field	Detail
Prepared for	Nestle / Procurement Scheduling Use Case
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Version	v3.0 - Complete process-flow replacement
Date	2026-07-17
Status	For review and sign-off
Supersedes	v2.0 workflow and scheduling sections

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Table of Contents

Document reading guide

Sections 5-7 contain the new process architecture and business logic. Sections 8 and 15 preserve and extend the existing registration and security requirements. Sections 9-12 define notifications, screens, reporting, and configuration. Sections 13-17 convert the process into implementable requirements and test criteria.

Section	Title
1	Executive Summary and Revision Scope
2	System Scope and Boundaries
3	Users, Roles and Responsibilities
4	Business Rules at a Glance
5	System Architecture
6	New End-to-End Process Flow
7	Detailed Scheduling, Confirmation and State Logic
8	Registration and Data Collection Fields
9	Notifications and Reminder Rules
10	Dashboard and Mobile App Requirements
11	Reporting and Monthly PDF Export
12	Configuration and Administration
13	Functional Requirements
14	Data Model and Records
15	Security Analysis
16	Non-Functional Requirements
17	Acceptance Criteria
18	Assumptions, Risks and Open Points

1. Executive Summary and Revision Scope

Go See is a standalone web and mobile platform that coordinates site visits between Procurement, Engineers, and Suppliers after a PR has been raised in the existing Nestle system. Procurement manually creates a Go See job, assigns the engineer, filters and selects suppliers, and starts the scheduling cycle.

The new workflow uses one engineer-selected visit time within a rolling three-day window. Suppliers only respond Available or Not available. A time is automatically confirmed when at least three suppliers accept. If the threshold is not reached by the response cutoff, the system applies a fixed decision rule: two acceptances require a Procurement override with justification; zero or one acceptance automatically returns the job to the engineer for a new time.

After the visit, the engineer records which confirmed suppliers attended and closes the visit cycle. Procurement then has a configurable closure period, initially three days, to close the job or recirculate it. Recirculation creates a new cycle under the same PR and excludes suppliers who already attended, while allowing previously unavailable, non-responsive, no-show, and newly selected suppliers.

Critical workflow replacement

The previous multi-slot matching and supplier alternative-time proposal flow is removed. The new process is based on one engineer-selected time, supplier availability confirmations, a minimum confirmation threshold, and a controlled override / reschedule rule.

1.1 Main additions in v3.0

- Searchable engineer selection during job creation.
- Category and sub-category filtering with Select All, individual selection, and search within filtered suppliers.
- Existing supplier-visit suggestions inside the engineer's active three-day timeline.
- Available / Not available supplier responses for one selected visit time.
- Automatic confirmation at three accepted suppliers and a 24-hour response cutoff rule.
- Procurement override with mandatory justification when exactly two suppliers accept.
- Automatic engineer rescheduling when zero or one supplier accepts.
- Engineer attendance verification, repeated closure reminders, and Procurement closure SLA.
- Recirculation that excludes suppliers who attended and preserves every cycle for audit and reporting.
- Monthly PDF export and configurable thresholds, deadlines, slot rules, and reminder frequencies.

2. System Scope and Boundaries

Area	Included in Go See	Boundary / Not Included
PR initiation	Store PR number and description entered by Procurement.	PR creation, approval, and ERP workflow remain in the Nestle system.
Job assignment	Assign an active engineer and one or more active suppliers using category / sub-category filters.	No automatic procurement approval or supplier commercial ranking.
Scheduling	One time selected by engineer within the active three-day window; show supplier visit suggestions.	No supplier-proposed alternative time in the new flow.
Supplier response	Available / Not available, threshold counting, cutoff decision, override, and reschedule.	The system cannot physically force attendance; it records participation outcomes.
Visit attendance	Engineer records attendance for confirmed suppliers and closes the visit cycle.	Detailed work completion certification is outside scope unless added later.
Procurement outcome	Close the job or recirculate to eligible suppliers; store justification and history.	Quotation evaluation and commercial award decision remain outside Go See.
Notifications	OTP, job invitations, confirmation, reminders, reschedule and escalation SMS / app messages.	Email and Microsoft Teams integration are excluded from the initial scope.
Reporting	Monthly PDF exports with PR, job, schedule, responses and attendance.	Financial analytics and ERP purchase-order reporting are outside the initial scope.
Internal systems	Standalone VPS / apps / database with manually entered PR details.	No direct connection to Nestle ERP, SAP, Active Directory or internal file systems.

Operational boundary

Go See starts only after Procurement receives or reviews the PR and creates the job. The platform stores the PR reference but does not replace the existing PR or ERP process.

3. Users, Roles and Responsibilities

Role	Platform	Main Responsibilities	Key Permissions
Procurement Team	Web dashboard	Create jobs, search engineers, filter/select suppliers, monitor responses, approve two-supplier override, review attendance, close or recirculate jobs, export reports.	Assigned procurement functions; all high-impact actions are audited.
Procurement Manager	Web dashboard / SMS	Review overdue closure escalations, monitor exceptions, approve policy where required.	Manager-level oversight and escalation visibility.
Engineer	Mobile app	View assigned jobs, review supplier-visit suggestions, select one visit time, reschedule when required, mark supplier attendance, close visit cycle.	Can act only on jobs assigned to the engineer.
Supplier	Mobile app	View assigned invitations, respond Available / Not available, view confirmed time and reminders.	Can view only jobs where the supplier is selected.
System Admin	Dashboard / backend	Manage configuration, reminder rules, OTP/SMS settings, uptime, audit logs and support.	Technical administration; no procurement decision authority unless separately assigned.

Access principle

Role-based access control is enforced on the server. Every job, response, attendance, closure and report request must also pass object-level authorization so users cannot access another person's records by changing an ID.

4. Business Rules at a Glance

ID	Rule	Required Behaviour
BR-01	PR source	PR is raised outside Go See. Procurement manually enters the PR number and description.
BR-02	Single selected time	The engineer selects one visit time, not multiple alternatives.
BR-03	Three-day window	The selected time must fall inside the configured next-three-day window and respect the minimum lead time.
BR-04	Supplier invitation	Every supplier selected by Procurement receives the same visit time and responds Available or Not available.
BR-05	Automatic threshold	When three suppliers respond Available, the time is immediately confirmed and locked.
BR-06	Mandatory confirmed time	After confirmation, the visit time does not change because of later responses. All selected suppliers are notified that the time is confirmed and attendance is required; each supplier must still record a response.
BR-07	Cutoff rule	At 24 hours before the selected time: exactly two available responses require Procurement override; zero or one automatically requires engineer rescheduling.
BR-08	Override evidence	Procurement must enter a justification before proceeding with two suppliers. Decision and reason are audited.
BR-09	Attendance closure	After the visit, the engineer records attendance for the confirmed supplier list and closes the cycle.
BR-10	Engineer reminders	If the engineer has not closed the cycle after the configured grace period, reminder SMS messages repeat at the configured frequency.
BR-11	Procurement closure SLA	Procurement closes or recirculates the job within three days after engineer closure. Overdue jobs trigger a Procurement Manager SMS.
BR-12	Recirculation exclusion	Suppliers who attended the previous cycle cannot be selected again for the same PR cycle. Previous unavailable, no-response, no-show and new suppliers remain eligible.
BR-13	Cycle history	Every recirculation creates a new numbered cycle while preserving earlier responses, attendance and decisions.

5. System Architecture

The proposed architecture remains standalone but is expanded to support time-based rules, response cutoffs, repeating reminders, attendance closure, recirculation and reporting. A background scheduler / queue is required because several actions happen automatically at future times rather than during a user request.

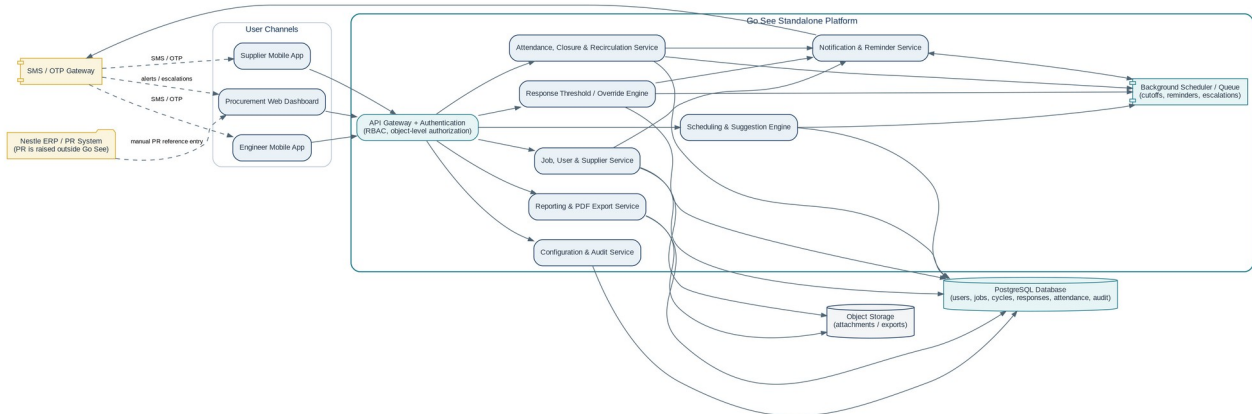


Figure 1: High-level standalone architecture and boundary with the Nestle PR system.

5.1 Suggested Technical Components

Component	Responsibility	Example Implementation
Procurement Dashboard	Job creation, engineer search, supplier filtering/search/selection, monitoring, override, attendance view, closure, recirculation, reports and configuration.	Next.js / React web application
Engineer Mobile App	OTP login, job inbox, three-day timeline, suggestion view, single-time selection, reschedule, attendance marking and engineer closure.	React Native / Expo
Supplier Mobile App	OTP login, job inbox, Available / Not available response, confirmed visit view and reminders.	React Native / Expo
API & Auth Layer	Authentication, RBAC, object-level access, validation, idempotency and rate limiting.	Node.js / NestJS / Express
Job & Master Data Service	Users, suppliers, categories, sub-categories, job setup, attachments and active/inactive rules.	Backend service/module
Scheduling & Suggestion Engine	Three-day validation, minimum lead time, existing-visit lookup and suggestion ordering.	Backend service/module
Threshold & Override Engine	Counts available responses, locks at three, evaluates cutoff, requests override and records justification.	Backend service/module
Attendance / Closure Service	Engineer attendance, visit closure, Procurement SLA, manager escalation and final closure.	Backend service/module
Recirculation Service	Creates new cycle, builds eligible suppliers, excludes attended suppliers and preserves history.	Backend service/module
Notification / Reminder Service	OTP, invitation, confirmation, reschedule, repeated reminders and escalation messaging.	Queue workers + SMS provider
Reporting Service	Monthly filtering, PDF generation and export audit trail.	Server-side reporting module
Database & Object Storage	Relational records, audit trail, attachments and generated reports.	PostgreSQL / Supabase + object storage

5.2 Event-Driven Processing

- When an engineer selects a time, the system schedules a response-cutoff event for 24 hours before the visit.
- When the third Available response arrives, the threshold event is completed early and the time is locked immediately.
- At the cutoff, the background worker evaluates the count and creates an override or reschedule task.
- After the visit grace period, a worker checks engineer closure and repeats reminders until closure.
- After engineer closure, a Procurement closure deadline is created; overdue jobs trigger manager escalation.
- All scheduled actions must be idempotent so duplicate queue delivery cannot send conflicting confirmations or change a locked appointment.

5.3 Timer and Event Catalogue

Event	Created When	Due Time	Action
Response cutoff	Engineer time selected	Selected time minus configured cutoff	Auto-confirm / override / reschedule decision
Supplier reminder	Supplier invitation sent	Configured delays before cutoff	Send reminder only while response is Pending
Engineer closure check	Visit time reached	Visit time plus grace period	Start repeating closure reminders
Procurement closure deadline	Engineer closes cycle	Engineer close time plus closure SLA	Mark overdue and escalate manager
Report generation	User requests export	Immediate / queued	Create PDF and secure download record

5.4 Worker Reliability Controls

- Use a durable queue so scheduled decisions survive server restarts.
- Use idempotency keys based on cycle and event type.
- Move repeatedly failing tasks to a dead-letter queue and alert System Admin.
- Store last attempt, next attempt and final outcome for every automated action.
- Use database transactions / row version checks when confirming or changing a cycle state.

6. New End-to-End Process Flow

The following diagram is the authoritative replacement for the previous scheduling process. Detailed phase descriptions follow on portrait pages.

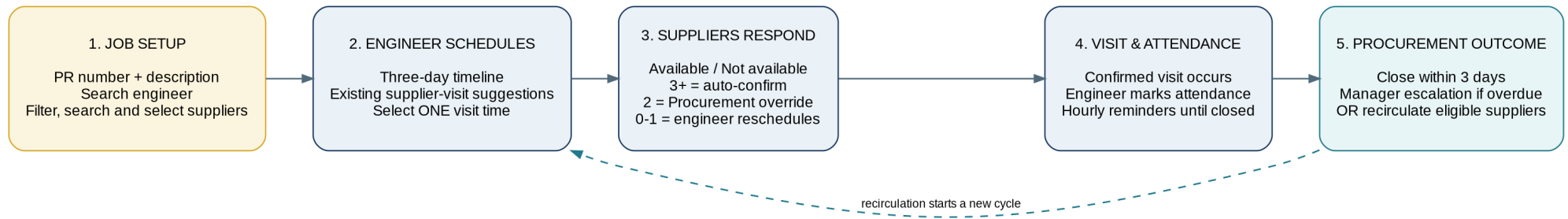


Figure 2: Complete Go See process from PR reference to final closure or recirculation.

6.1 Phase A - PR Reference and Job Creation

1. An engineer raises the PR in the existing Nestle ERP / PR process.
2. Procurement opens the Go See dashboard and creates a job using the PR number and PR description.
3. Procurement selects the assigned engineer using a searchable Engineer Name field. Only active engineers appear.
4. Procurement selects a main supplier category and sub-category. The system displays only active suppliers matching both filters.
5. Procurement may use Select All for the filtered result or select suppliers one by one. A search box must search only within the filtered result.
6. Procurement reviews the selected supplier list and creates the job.
7. The system sends an SMS / app notification to the assigned engineer: the PR job has been created and a visit time must be selected.

6.2 Phase B - Engineer Time Selection and Suggestions

8. When the engineer opens the job, the app displays a calendar / timeline similar to a meeting scheduling assistant.
9. The selectable period runs from the allowed minimum lead time up to the end of the configured three-day window. Invalid times are disabled.
10. Before the engineer selects a time, the system checks whether any selected supplier already has another pending or confirmed Go See visit in the window.
11. Suggested blocks show that a selected supplier is already expected on site and recommend a nearby time before or after that visit. Other job or PR details must not be exposed.
12. The engineer selects one visit time and confirms it.

6.3 Phase C - Supplier Response and Time Confirmation

13. All selected suppliers receive an SMS and app invitation containing the PR reference, selected time, and instruction to respond in the Go See app.
14. Each supplier chooses Available or Not available. Responses remain editable only until the time is locked or the response cutoff is reached.
15. As soon as three suppliers respond Available, the system confirms and locks the selected time.
16. After locking, every selected supplier receives a second notice: the visit time is confirmed and attendance is required. Suppliers who have not responded must still open the app and record their response.
17. At 24 hours before the visit, if only two suppliers are Available, Procurement receives an Override Required alert and may proceed only after entering a justification.
18. At the cutoff, if zero or one supplier is Available, the system rejects that time and sends the engineer a reschedule request. The engineer selects another time and the supplier-response cycle repeats.

6.4 Phase D - Visit Attendance and Engineer Closure

19. The confirmed Go See visit takes place at the selected time.
20. After the configured grace period, initially one hour, the system checks whether the engineer has closed the visit.
21. The engineer opens the assigned job and sees the confirmed supplier list.
22. The engineer can select all suppliers or mark them one by one as Attended or Did not attend, then confirms whether the visit occurred.
23. The engineer closes the visit cycle. Procurement can immediately see final attendance and non-attendance.
24. If the cycle is not closed, an SMS reminder repeats at the configured frequency, initially every one hour, until closure.

6.5 Phase E - Procurement Closure or Recirculation

25. After engineer closure, Procurement has a configurable period, initially three days, to complete the job outcome.
26. Procurement may close the job when the process is complete.
27. If quotations or participation are not sufficient, Procurement may recirculate before final closure.
28. Recirculation creates a new numbered cycle under the same PR. Suppliers who attended are excluded. Previous suppliers who were unavailable, did not respond, or did not attend remain eligible, and new matching suppliers can be added.
29. The assigned engineer receives a new SMS and selects a new time. The same threshold, attendance and closure rules repeat until Procurement finally closes or cancels the job.
30. If Procurement does not close or recirculate within the closure period, the Procurement Manager receives an overdue SMS alert.

7. Detailed Scheduling, Confirmation and State Logic

7.1 Three-Day Window and Minimum Lead Time

ID	Rule	Expected Behaviour
SCH-01	Window start	The first selectable time starts after the configured minimum lead time. Recommended initial value: 24 hours.
SCH-02	Window end	The maximum selectable time is the configured rolling window end. Recommended initial value: 72 hours from engineer selection.
SCH-03	Slot duration	Available times follow the configured slot increment / minimum duration. Recommended initial value: 30 minutes, subject to sign-off.
SCH-04	Working hours	Only configured site working hours and permitted days are selectable.
SCH-05	Locked time	Once confirmed, the time cannot be changed by supplier responses. Only authorised cancellation / recirculation can end the cycle.
SCH-06	Reschedule	A failed cutoff returns the same cycle to Awaiting Engineer Time or creates a new attempt based on implementation; all prior responses remain auditable.

7.2 Existing Supplier Visit Suggestion Logic

Condition	System Action
Selected supplier has a confirmed visit in the window	Show the confirmed block first and label it "Supplier already scheduled". Suggest a configurable gap before or after.
Selected supplier has a pending visit in the window	Show as a softer pending suggestion below confirmed suggestions.
More than one selected supplier shares a time block	Rank the block higher because it can consolidate multiple supplier visits.
No relevant visit exists	Show the normal blank timeline.
Engineer cannot use a suggestion	Allow any valid time inside the configured window.
Privacy rule	Do not show another engineer name, PR description, commercial data, or unrelated job details.

7.3 Supplier Confirmation Threshold

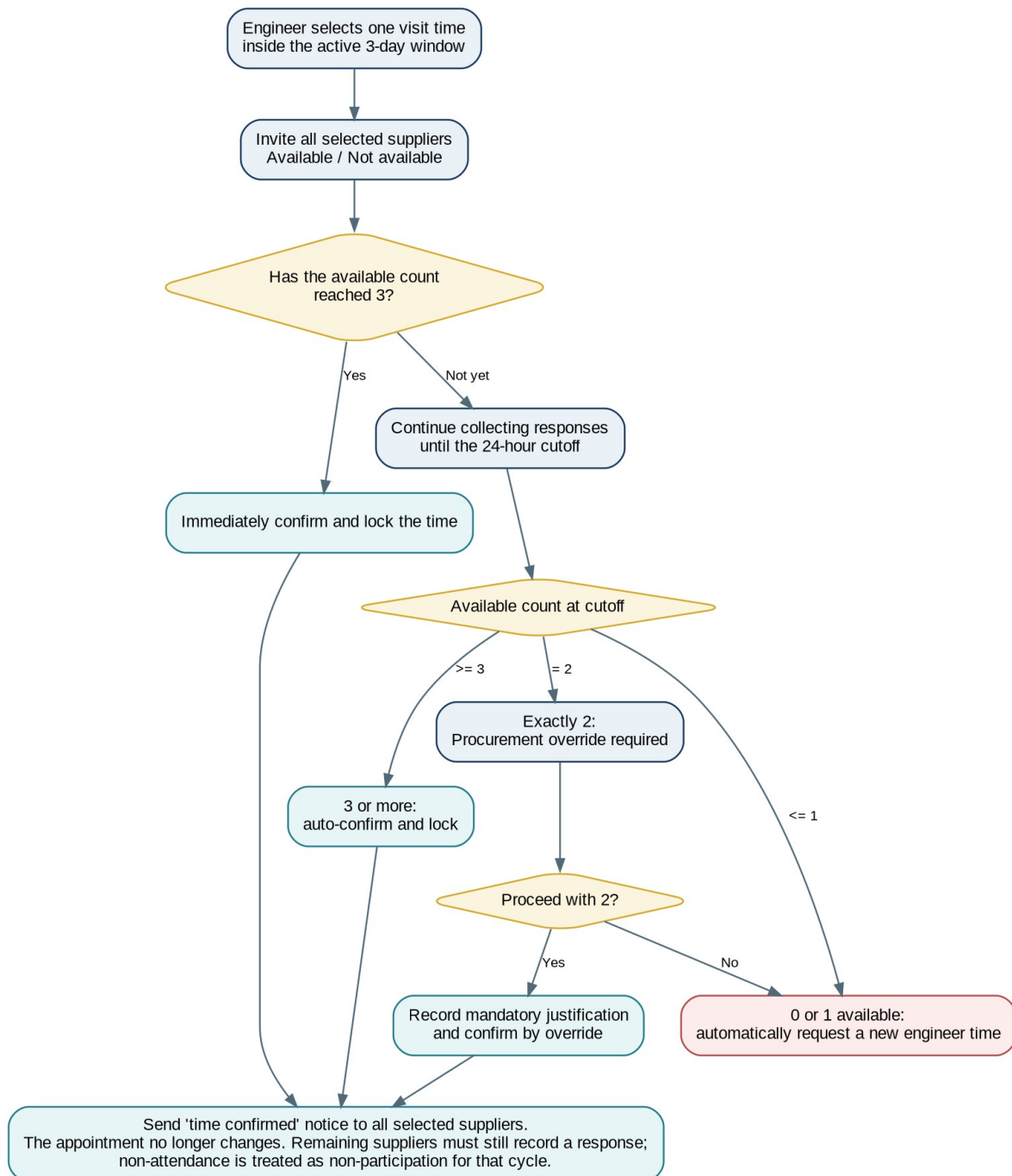


Figure 3: Response threshold, cutoff, override and rescheduling logic.

Available Count / Scenario	Evaluation Time	Decision	Follow-up
3 or more Available before cutoff	Immediately	Auto-confirm and lock selected time.	All selected suppliers, engineer and Procurement are notified.
3 or more Available at cutoff	At cutoff	Auto-confirm and lock selected time.	Same as above.
Exactly 2 Available at cutoff	At cutoff	Set Override Required.	Procurement must choose Proceed or Reschedule. Proceed requires justification.
0 or 1 Available at cutoff	At cutoff	Set Reschedule Required.	Notify engineer to select a new time; notify Procurement.
Later supplier declines after time locked	After confirmation	Do not unlock or change time.	Record response; attendance / non-participation is reported.
Supplier gives no response	Until visit	Keep No Response status.	Continue configured reminders; report separately.

7.4 Attendance and Closure Logic

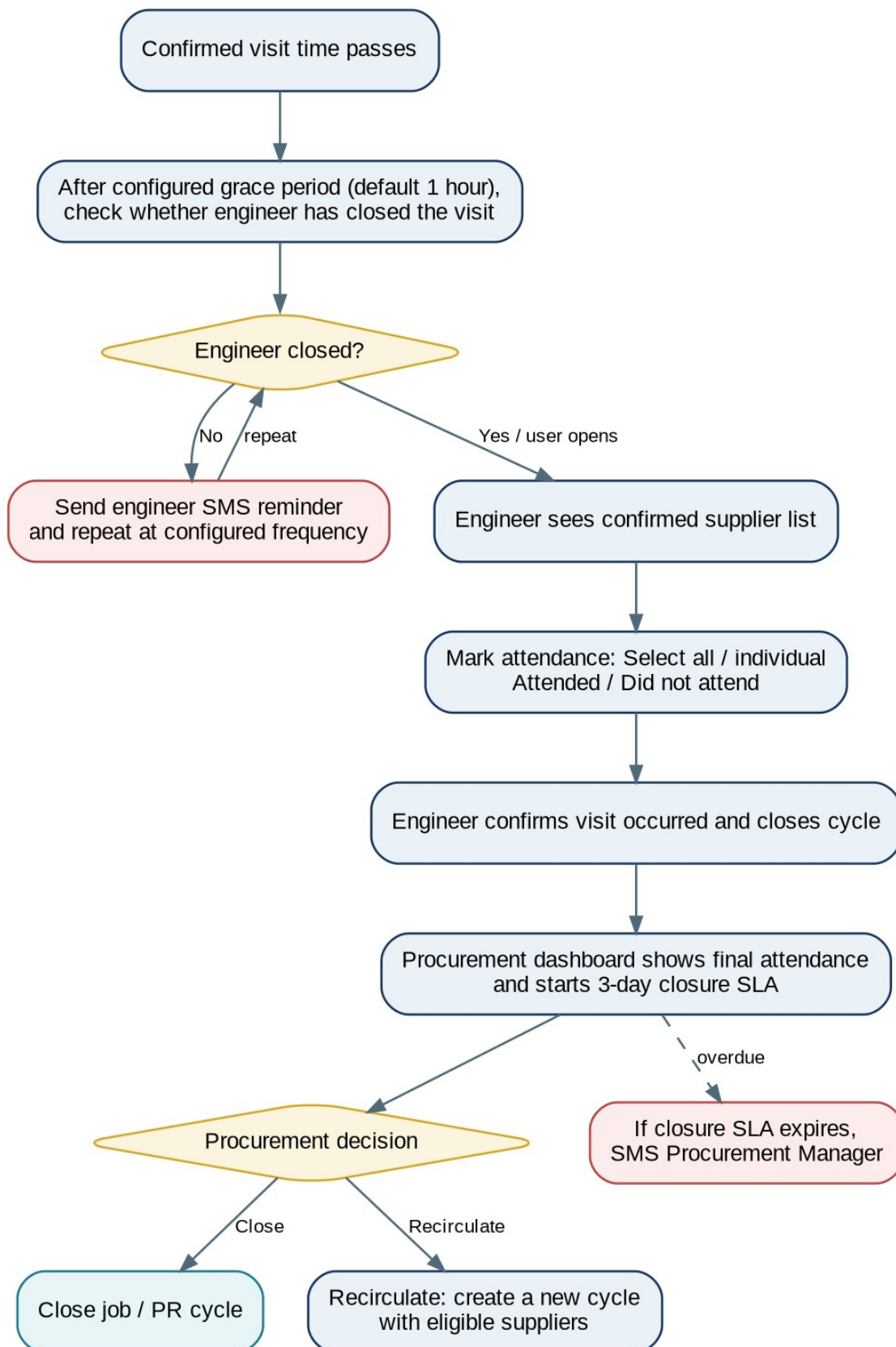


Figure 4: Engineer attendance closure and Procurement close / recirculate decision.

7.5 Recirculation Eligibility

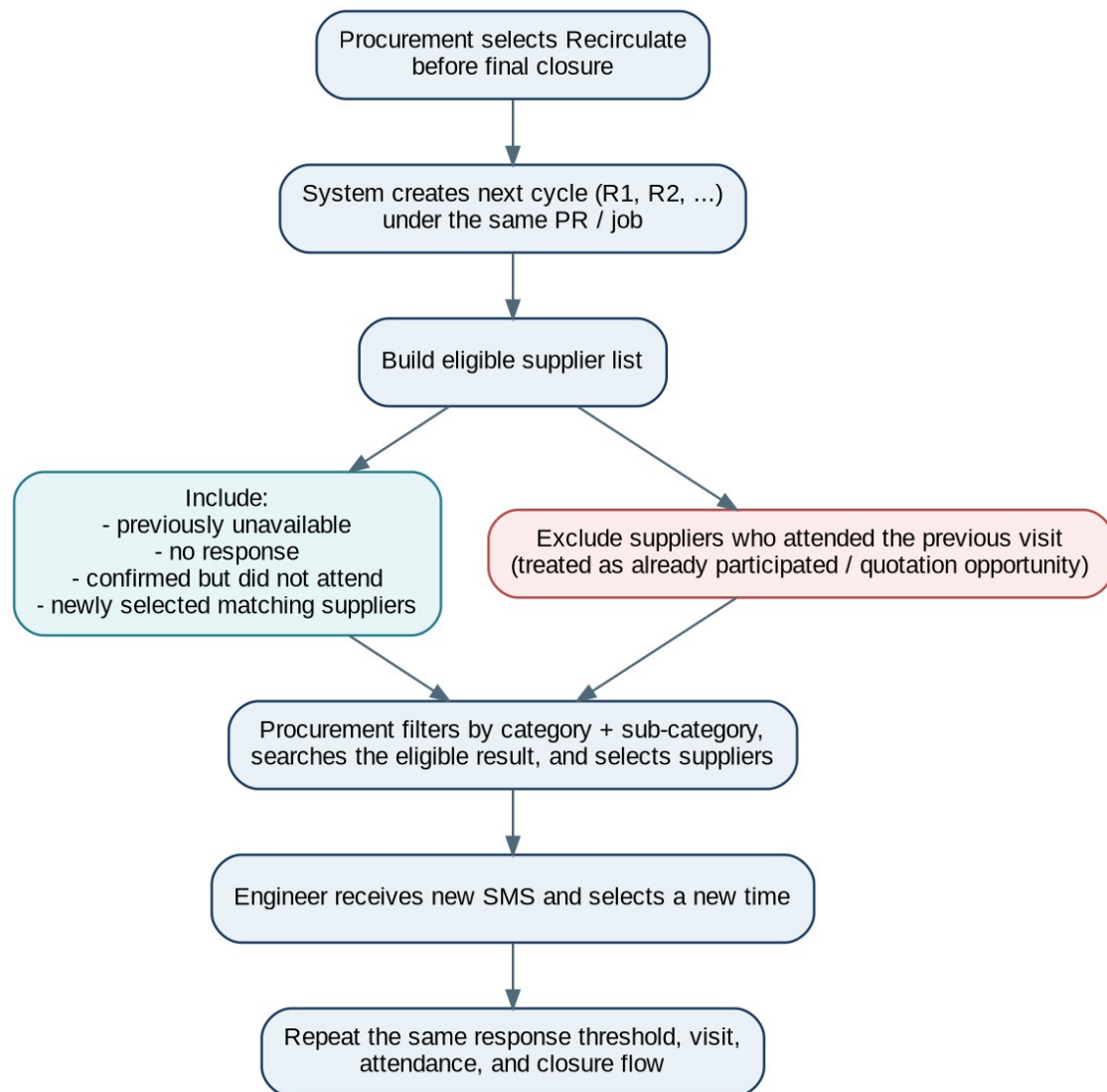


Figure 5: Supplier eligibility when a job is recirculated.

7.6 Job and Cycle Status Model

The parent Job remains under the same PR. Each recirculation creates a new Job Cycle so prior responses, attendance and override decisions are never overwritten.

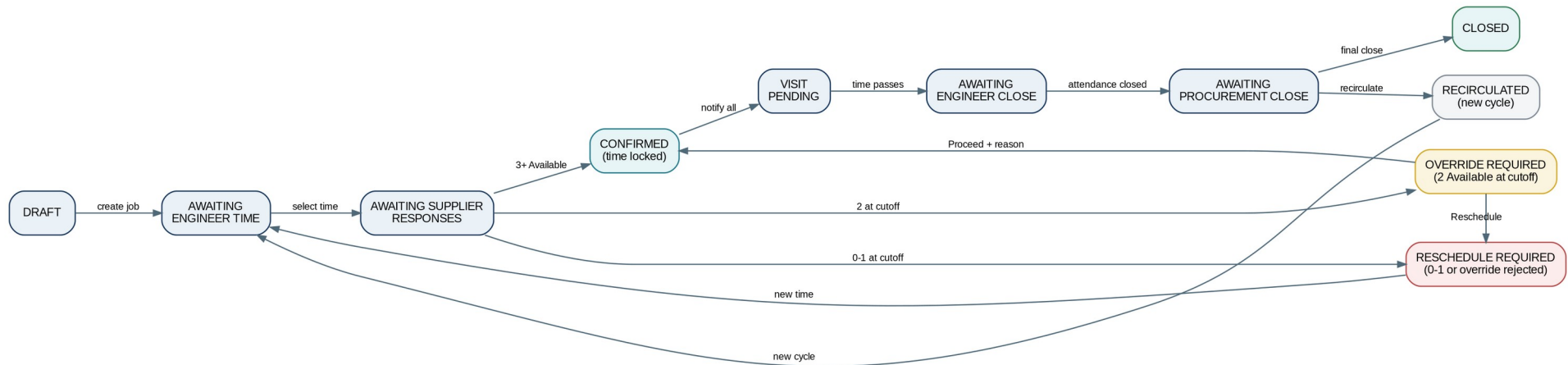


Figure 6: Recommended job-cycle status transitions.

Status	Meaning
DRAFT	Procurement is preparing the job; no notifications sent.
AWAITING_ENGINEER_TIME	Job created or reschedule requested; engineer must select one time.
AWAITING_SUPPLIER_RESPONSES	Time selected; supplier invitations sent; threshold not yet completed.
OVERRIDE_REQUIRED	Exactly two Available at cutoff; Procurement decision and justification required.
RESCHEDULE_REQUIRED	Zero or one Available at cutoff, or Procurement rejected the two-supplier override.
CONFIRMED	Time locked automatically or by approved override.
VISIT_PENDING	Confirmed time has not yet passed.
AWAITING_ENGINEER_CLOSE	Visit time passed; attendance and visit confirmation are pending.
AWAITING_PROCUREMENT_CLOSE	Engineer closed cycle; Procurement must close or recirculate.
RECIRCULATED	Current cycle ended and a new numbered cycle has started.
CLOSED	Procurement completed final closure.
CANCELLED	Authorised user cancelled the job with a reason.

8. Registration and Data Collection Fields

The registration fields below are retained from the approved v2.0 document. New scheduling-cycle, response, attendance, override and closure fields are added so the revised process can be implemented and audited.

8.1 Engineer Registration Fields

Field	Type	Required	Notes / Validation
Engineer Name	Text	Yes	Full name. Searchable during job creation.
Mobile Number	Phone	Yes	Unique; used for OTP login and SMS.
Email	Email	No	Optional record / future notification channel.
Department / Section	Dropdown / Text	Yes	e.g. Utilities, Production, Maintenance.
Designation	Text	No	e.g. Maintenance Engineer, Executive.
Preferred Language	Dropdown	No	English initially; Sinhala / Tamil configurable.
Status	Toggle	Yes	Inactive engineers cannot be assigned.
Login Method	System	-	Mobile number + OTP; no mobile password stored.

8.2 Supplier Registration Fields

Field	Type	Required	Notes / Validation
Supplier Company Name	Text	Yes	Unique company / vendor name.
Contact Person Name	Text	Yes	Primary app and coordination contact.
Mobile Number	Phone	Yes	Unique; used for OTP and SMS.
Email	Email	No	Optional record.
Supplier Category	Multi-select	Yes	Supplier may belong to one or more categories; inline category creation allowed.
Supplier Sub-Category / Spend Tier	Dropdown	Yes	Configurable tier; inline creation allowed.
Address	Multi-line Text	No	Supplier address / operating location.
Availability Contact Number	Phone	No	Secondary scheduling number.
Preferred Language	Dropdown	No	English / Sinhala / Tamil.
Status	Toggle	Yes	Inactive suppliers are hidden from selection.
Login Method	System	-	Mobile number + OTP.

8.3 Supplier Category and Sub-Category

Item	Example	Behaviour
Main Category	Civil, Mechanical, Electrical, Painting, etc.	Configurable; create, edit, activate/deactivate; a supplier can have multiple.
Sub-Category / Spend Tier	Low-Value / Petty, Standard, High-Value / Strategic	Configurable; used with main category to narrow suppliers.
Inline Creation	During supplier registration or job creation	Restricted to authorised users; check for duplicate names before save.

8.4 Job Creation Fields (Procurement)

Field	Type	Required	Notes / Validation
PR Number	Text	Yes	Reference from existing Nestle system; searchable and preferably unique per active job.
PR Description / Job Description	Multi-line Text	Yes	Short operational description of requirement.
Job Title	Text	No	May be generated from PR description or entered manually.
Engineer Name	Searchable Select	Yes	Search active engineers by name; display department / designation for disambiguation.
Site / Location	Dropdown / Text	Yes	Visit location.
Priority	Dropdown	Yes	Normal / Urgent or configurable list.
Main Category Filter	Dropdown	Yes	Filters active suppliers.
Sub-Category Filter	Dropdown	Yes	Further filters suppliers.
Supplier Search	Search Box	No	Searches only within the current filtered result.
Select All Filtered Suppliers	Checkbox / Button	No	Select all currently displayed matching suppliers.
Selected Suppliers	Multi-select	Yes	Select individually or all; removable before creation.
Expected Visit Requirement	Dropdown / Text	No	e.g. inspection, site visit, quotation.
Attachments / Notes	File / Text	No	Supporting information, subject to file controls.

8.5 Scheduling Cycle, Response and Closure Fields

Field	Type	Purpose
Cycle Number	System	R0, R1, R2... under the same job / PR.
Engineer Selected Time	DateTime	One time within the allowed window.
Response Cutoff	DateTime	Calculated from selected time and cutoff configuration.
Supplier Response	Enum	Pending / Available / Not Available / No Response.
Response Timestamp	DateTime	Audit when supplier responded or changed response.
Confirmation Source	Enum	Automatic Threshold / Procurement Override.
Override Decision	Enum	Proceed / Reschedule.
Override Justification	Multi-line Text	Mandatory when Proceed with exactly two suppliers.
Visit Happened	Yes / No	Engineer confirmation.
Attendance Status	Enum per supplier	Attended / Did Not Attend / Not Marked.
Engineer Closed At	DateTime	Starts Procurement closure SLA.
Procurement Action	Enum	Close / Recirculate / Cancel.
Procurement Reason	Text	Mandatory for recirculate or cancel; optional for normal close.

9. Notifications and Reminder Rules

ID	Recipient	Trigger	Message Intent / Example
N-01	Engineer	Job created	PR [number] has been created in Go See. Open the app and select one visit time.
N-02	Supplier	Invitation	PR [number] - Engineer selected [date/time]. Open Go See and confirm Available or Not available.
N-03	Supplier	Time confirmed	The visit time for PR [number] is confirmed for [date/time]. Attendance is required. Open the app and record your confirmation.
N-04	Procurement	Override required	Only two suppliers are Available for PR [number]. Open the dashboard to Proceed with justification or request rescheduling.
N-05	Engineer	Reschedule required	The selected time for PR [number] did not receive enough supplier confirmations. Please select another time.
N-06	Engineer	Visit closure reminder	Please close PR [number] in Go See and mark which suppliers attended.
N-07	Procurement	Engineer closed	Attendance is available for PR [number]. Close or recirculate the job within the configured period.
N-08	Procurement Manager	Closure overdue	PR [number] has passed the Procurement closure deadline. Please review and close or recirculate.
N-09	Engineer	Recirculated	PR [number] has been recirculated. Open Go See and select a new visit time.
N-10	Supplier	Response reminder	Your response is pending for PR [number] at [date/time]. Open Go See before the cutoff.

9.1 Reminder Behaviour

Reminder	Starts	Recommended Frequency	Stops
Supplier response	After invitation and before cutoff	Configurable; recommended initial reminder at 6 hours after invite and again 2 hours before cutoff.	Stop when supplier responds or cycle leaves Awaiting Supplier Responses.
Engineer reschedule	After Reschedule Required	Configurable; recommended every 6 hours during working hours.	Stop when a new time is selected or job is cancelled.
Engineer visit closure	After visit grace period	Default every 1 hour.	Repeat until engineer closes the cycle.
Procurement closure	After engineer closure	Dashboard aging; optional daily reminder.	Stop when close / recirculate / cancel is completed.
Manager escalation	When Procurement SLA expires	Immediate SMS, then configurable repeat.	Stop when Procurement completes an action.

SMS content control

Messages should contain only the minimum operational details. Do not include sensitive attachments, quotation values, or unrelated supplier information in SMS.

10. Dashboard and Mobile App Requirements

10.1 Procurement Dashboard

Screen / Module	Main Elements
Main Dashboard	Counts for Awaiting Engineer, Awaiting Suppliers, Override Required, Confirmed Today, Awaiting Engineer Close, Awaiting Procurement Close, Overdue, Recirculated and Closed.
Job Search & Filters	PR number, description, engineer, category, sub-category, supplier, cycle status, date range, overdue flag and recirculation count.
Create Job	PR number, description, searchable engineer, location, priority, category/sub-category filters, supplier result search, Select All, individual selection and selected supplier summary.
Job Detail	Parent PR summary, current cycle, selected time, response count, per-supplier response, cutoff, confirmation source, attendance, notifications and complete history.
Override Panel	Available count, supplier names/statuses, Proceed / Reschedule buttons and mandatory justification for Proceed.
Attendance View	Confirmed suppliers with Attended / Did Not Attend, engineer close timestamp and reminder history.
Close / Recirculate	Close, Recirculate or Cancel; recirculation supplier eligibility; reason field; new supplier filters.
User Management	Create/update/deactivate engineer and supplier accounts; search and filter.
Category Management	Create/edit/activate/deactivate categories and sub-categories; duplicate prevention.
Reports	Monthly and custom date filters; preview and export PDF.
Configuration	Thresholds, window, lead time, working hours, reminders, SLA and notification templates.

10.2 Engineer Mobile App

Screen / Module	Main Elements
OTP Login	Mobile number, OTP, resend cooldown and language.
Job Inbox	Assigned jobs grouped by action required: Select Time, Reschedule, Visit Pending, Close Visit, Completed.
Job Detail	PR reference, description, location, priority and selected supplier list.
Time Scheduler	Three-day timeline, working hours, disabled invalid slots, confirmed / pending supplier-visit suggestions and single-time selection.
Reschedule Screen	Reason for failed time and new timeline selection.
Attendance Screen	Confirmed supplier list; Select All; individual Attended / Did Not Attend; visit happened confirmation.
Closure Confirmation	Final attendance summary and irreversible Close Cycle confirmation.

10.3 Supplier Mobile App

Screen / Module	Main Elements
OTP Login	Mobile number, OTP and language preference.
Job Inbox	Pending response, confirmed visits and completed invitations.
Invitation Detail	PR number, description, engineer / site contact, location, selected date/time and response cutoff.
Response	Available and Not available actions; clear confirmation and timestamp.
Confirmed Visit	Locked date/time, location, attendance requirement and current response status.
History	Past invitations and participation outcomes visible only to that supplier.

10.4 Common Interaction and Status Rules

Rule	Required Behaviour
Action Required Banner	Every user inbox clearly shows the next action and deadline.
Status Colour + Text	Never rely on colour alone; show status text and icon.
Confirmation Summary	Before irreversible actions, show PR, time, supplier count and consequence.
Offline / Weak Network	Prevent duplicate submission; show retry state and keep the app inbox as source of truth.
History	Show cycle timeline with invitations, responses, confirmation, attendance and closure events.
Accessibility	Readable contrast, scalable text, large touch targets and keyboard support on dashboard.

11. Reporting and Monthly PDF Export

The Procurement dashboard shall export a readable monthly PDF report. The report must preserve cycle history so one PR with multiple recirculations can be reviewed without losing the earlier supplier outcomes.

Section	Required Data
Report Header	Period, generated date/time, generated by, active filters and summary counts.
PR / Job	PR number, description, site, priority, engineer, category and sub-category.
Cycle	Cycle number, created date, selected time, response cutoff, confirmation source and status.
Supplier Responses	Selected supplier, Available / Not Available / No Response, response timestamp.
Confirmation	Automatic threshold or Procurement override, available count and override justification.
Attendance	Supplier attended, did not attend, not marked; engineer close timestamp.
Procurement Outcome	Closed, recirculated or cancelled; action date, reason and responsible user.
Overdue Indicators	Engineer closure delay, Procurement closure delay and manager escalation.

11.1 Filters and Summary Metrics

- Month / custom date range, PR number, engineer, site, category, sub-category, supplier, job status and cycle status.
- Jobs created, visits confirmed, automatic confirmations, override confirmations, reschedules, recirculations and final closures.
- Supplier response rate, attendance rate, no-response count, no-show count and average time to confirmation.
- Engineer closure compliance and Procurement closure SLA compliance.
- PDF is the mandatory export. CSV / Excel may be added as a useful secondary export without changing the PDF requirement.

Reporting classification

For this workflow, an attended supplier is treated as having participated in the Go See quotation opportunity. The system does not evaluate quotation price or award decisions.

12. Configuration and Administration

Configuration Item	Recommended Initial Value	Purpose
Active Scheduling Window	72 hours / next 3 days	How far ahead an engineer may select.
Minimum Lead Time	24 hours	Prevents selection too close to the cutoff.
Slot Increment / Minimum Duration	30 minutes - proposed	Timeline granularity; requires business sign-off.
Working Days / Hours	Site-specific	Selectable time ranges.
Automatic Confirmation Threshold	3 suppliers	Locks the selected time.
Override Count	Exactly 2 suppliers	Triggers Procurement override.
Response Cutoff	24 hours before visit	Runs the final count decision.
Minimum Suppliers Invited	3 - proposed	Warn or require exception when fewer suppliers are selected.
Supplier Reminder Rules	Configurable	Delay, repeat frequency and stop condition.
Engineer Closure Grace Period	1 hour after visit	When closure reminders begin.
Engineer Closure Reminder Frequency	Every 1 hour	Repeats until cycle is closed.
Procurement Closure SLA	3 days after engineer close	Time to close / recirculate / cancel.
Manager Escalation Frequency	Immediate on breach; repeat configurable	Overdue control.
Suggestion Gap	30-60 minutes - proposed	Recommended before / after existing supplier visit.
OTP Validity / Resend / Attempts	5 min / 60 sec / 5 - proposed	Security configuration.
Languages	English / Sinhala / Tamil	App and SMS template availability.

12.1 Configuration Governance

- Only authorised System Admin / Procurement Admin roles may change business rules.
- Every configuration change stores old value, new value, actor, date/time and reason.
- A rule change applies only to new cycles unless an authorised migration is explicitly approved.
- Dangerous combinations must be validated; for example, minimum lead time cannot be shorter than the response cutoff unless a defined urgent exception exists.
- Notification templates are versioned by language and message type.

13. Functional Requirements

ID	Requirement	Description	Priority
FR-01	OTP Login	Engineer and Supplier users shall authenticate using registered mobile number and OTP.	Must Have
FR-02	Dashboard Login	Procurement and System Admin shall use secure credential-based login.	Must Have
FR-03	Engineer Management	Create, update, search, activate and deactivate engineer accounts with retained fields.	Must Have
FR-04	Supplier Management	Create, update, search, activate and deactivate supplier accounts with category and sub-category mappings.	Must Have
FR-05	Category Management	Create, edit, activate/deactivate and prevent duplicate main categories and sub-categories.	Must Have
FR-06	Inline Category Creation	Authorised users may create a missing category or sub-category without leaving the form.	Must Have
FR-07	Job Creation	Procurement shall create a job using PR number, description, engineer, location, filters and selected suppliers.	Must Have
FR-08	Engineer Search	Engineer selection shall support search by Engineer Name and show identifying department/designation.	Must Have
FR-09	Supplier Filter	Supplier list shall filter by selected main category and sub-category and exclude inactive suppliers.	Must Have
FR-10	Search Filtered Suppliers	Procurement shall search only inside the current filtered supplier result.	Must Have
FR-11	Select All / Individual	Procurement shall select all filtered suppliers or select/remove suppliers individually.	Must Have
FR-12	Engineer Notification	After creation, the assigned engineer shall receive SMS/app notification to select a time.	Must Have
FR-13	Three-Day Scheduler	Engineer shall see a meeting-style timeline limited by configured window, lead time and working hours.	Must Have
FR-14	Existing Visit Suggestions	System shall show confirmed and pending visits involving selected suppliers as ranked suggestions without exposing unrelated job details.	Must Have
FR-15	Single Time Selection	Engineer shall select exactly one visit time per scheduling attempt.	Must Have
FR-16	Supplier Invitation	All selected suppliers shall receive the selected time through SMS/app notification.	Must Have
FR-17	Supplier Response	Supplier shall respond Available or Not available.	Must Have
FR-18	Response Audit	System shall store response value, timestamp, actor and changes.	Must Have
FR-19	Automatic Confirmation	The third Available response shall immediately confirm and lock the time.	Must Have
FR-20	Locked Time Notice	After confirmation, all selected suppliers, engineer and Procurement shall receive the confirmed-time notice.	Must Have
FR-21	Cutoff Evaluation	At configured cutoff, the system shall count Available responses and apply the 3 / 2 / 1-0 rule.	Must Have
FR-22	Two-Supplier Override	Exactly two Available responses shall create an override task for Procurement.	Must Have
FR-23	Override Justification	Proceeding with two suppliers shall require a non-empty justification and audit record.	Must Have
FR-24	Automatic Reschedule	Zero or one Available response at cutoff shall notify the engineer and set Reschedule Required.	Must Have

ID	Requirement	Description	Priority
FR-25	Supplier Response Reminders	Pending suppliers shall receive configured reminders until response, cutoff or cycle state change.	Should Have
FR-26	Attendance List	After the visit, engineer shall see the confirmed supplier list.	Must Have
FR-27	Select All Attendance	Engineer shall mark all suppliers or individual suppliers as Attended / Did Not Attend.	Must Have
FR-28	Engineer Cycle Closure	Engineer shall confirm visit happened and close the cycle.	Must Have
FR-29	Engineer Closure Reminders	System shall begin after configured grace period and repeat until engineer closure.	Must Have
FR-30	Procurement Attendance View	Dashboard shall show final supplier attendance immediately after engineer closure.	Must Have
FR-31	Procurement Closure SLA	System shall calculate the closure deadline from engineer close time.	Must Have
FR-32	Manager Escalation	System shall SMS the Procurement Manager when the closure SLA is exceeded.	Must Have
FR-33	Final Close	Procurement shall close the job / cycle with audit data.	Must Have
FR-34	Recirculate	Procurement shall create a new cycle under the same PR before final closure.	Must Have
FR-35	Recirculation Eligibility	System shall exclude attended suppliers and include eligible prior / new suppliers.	Must Have
FR-36	Recirculation Notification	Engineer shall receive a new time-selection notification for the new cycle.	Must Have
FR-37	Cycle History	All cycles, responses, overrides, attendance and closures shall remain visible and immutable in history.	Must Have
FR-38	Job Filtering	Dashboard shall filter jobs by PR, engineer, supplier, category, date, status and overdue condition.	Must Have
FR-39	Monthly PDF Export	Dashboard shall export monthly PDF with job, time, response, attendance and outcome details.	Must Have
FR-40	Configuration	Authorised users shall configure window, thresholds, cutoff, reminders, SLA, working hours and templates.	Must Have
FR-41	Audit Log	System shall audit account, configuration, job, time, response, override, attendance, recirculation and closure actions.	Must Have
FR-42	Trilingual Content	Supplier app and SMS shall support English, Sinhala and Tamil; other channels may phase in.	Should Have

14. Data Model and Records

The revised workflow requires a parent Job and child Job Cycle model. This prevents recirculation from overwriting the first visit attempt and makes monthly reporting and audit reliable.

14.1 Recommended Entity Relationship Overview

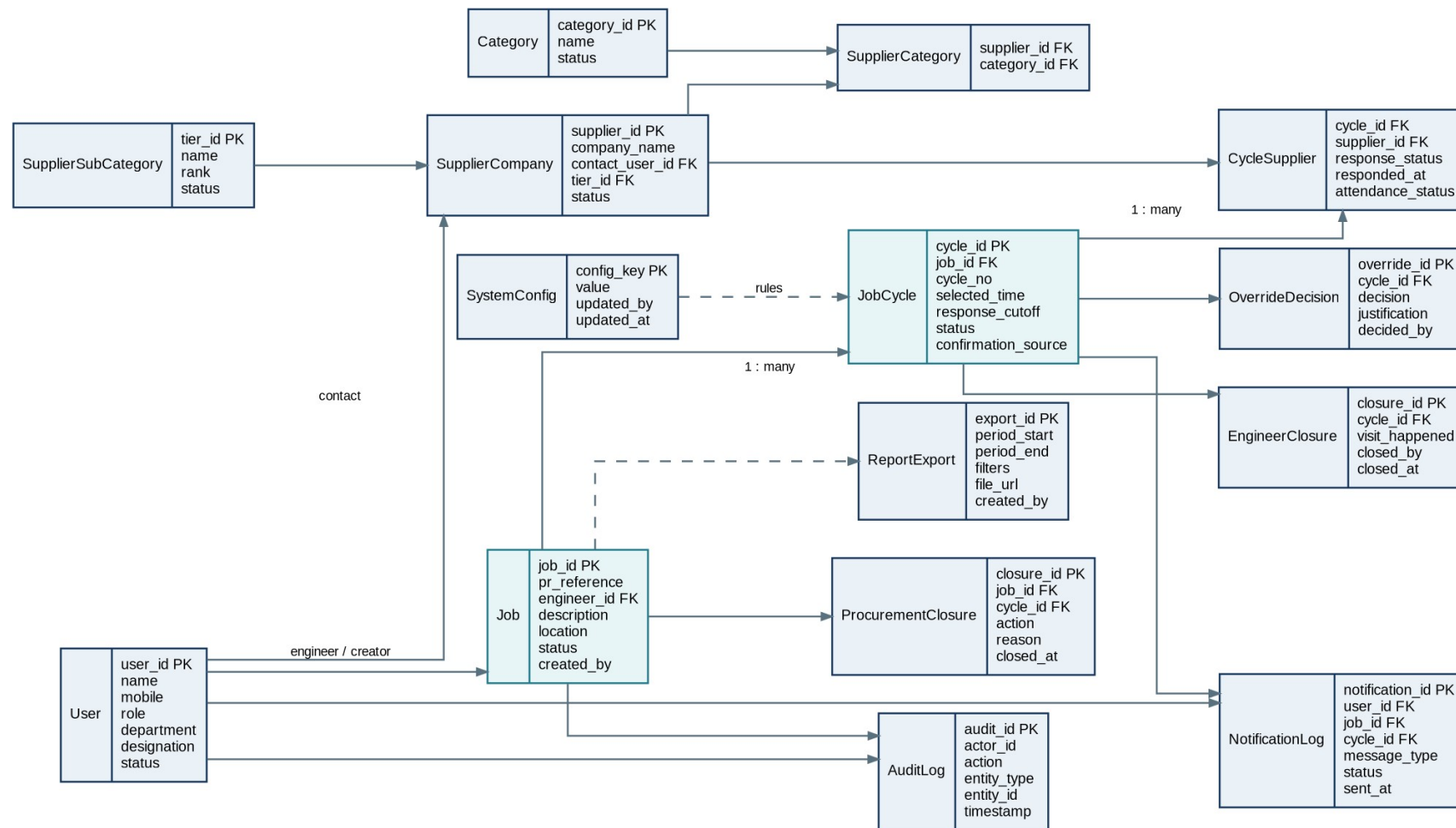


Figure 7: Simplified data model for job cycles, responses, attendance, closure and reporting.

Entity	Key Fields	Purpose
User	user_id, name, mobile, email, role, department, designation, preferred_language, status	Base record for engineer, supplier contact, Procurement and System Admin.
SupplierCompany	supplier_id, company_name, contact_user_id, address, availability_contact, tier_id, status	Supplier organisation and contact details.
Category	category_id, name, description, status, created_by	Main supplier category.
SupplierSubCategory	tier_id, name, description, rank, status	Spend tier / sub-category.
SupplierCategory	supplier_id, category_id	Many-to-many category mapping.
Job	job_id, pr_reference, description, engineer_id, location, priority, status, created_by, created_at	Parent record that remains across recirculations.
JobCycle	cycle_id, job_id, cycle_no, selected_time, response_cutoff, status, confirmation_source, created_at	One scheduling / visit attempt under the job.
CycleSupplier	cycle_id, supplier_id, invited_at, response_status, response_at, attendance_status	Supplier invitation, response and attendance for one cycle.
OverrideDecision	override_id, cycle_id, decision, justification, decided_by, decided_at	Procurement decision when exactly two suppliers are Available.
EngineerClosure	closure_id, cycle_id, visit_happened, attendance_completed, closed_by, closed_at	Engineer attendance and closure event.
ProcurementClosure	closure_id, job_id, cycle_id, action, reason, closed_by, closed_at	Close, recirculate or cancel decision.
SystemConfig	config_key, value, effective_from, updated_by, reason	Business and reminder settings.
NotificationLog	notification_id, user_id, job_id, cycle_id, channel, message_type, status, sent_at	SMS / app notification trace.
ReminderSchedule	reminder_id, cycle_id, reminder_type, due_at, recurrence, status	Future actions and repeating reminders.
ReportExport	export_id, period_start, period_end, filters, file_url, created_by, created_at	Generated PDF export history.
AuditLog	audit_id, actor_id, action, entity_type, entity_id, old_value, new_value, ip, timestamp	Accountability and troubleshooting.

14.2 Data Integrity Rules

- Only one active selected time exists for a cycle.
- A confirmed cycle cannot be returned to Awaiting Supplier Responses by a normal supplier action.
- One supplier has one current response per cycle; response history is stored in audit / event records.
- Attendance can be submitted only by the assigned engineer or an authorised Procurement override role.
- A new cycle cannot select a supplier whose previous attendance for the same job is Attended.
- Recirculation does not delete or update old CycleSupplier rows.
- Override Proceed requires justification; database and API validation must enforce it.
- Closure deadlines are calculated from immutable engineer close timestamps.

15. Security Analysis

The security requirements from the previous document remain applicable. The revised flow adds high-impact automated decisions, override justification, attendance records, manager escalation and report exports, so authorization and audit controls must cover these actions explicitly.

15.1 Security Objectives and Principles

- Confidentiality: supplier / engineer contact details and job data are visible only to authorised and assigned users.
- Integrity: selected time, threshold confirmation, override, attendance and closure cannot be altered by unauthorised users.
- Availability: OTP and SMS endpoints resist flooding and scheduled workers recover safely from failures.
- Least privilege: Procurement, Manager, Engineer, Supplier and System Admin functions are separated.
- Defence in depth: HTTPS, RBAC, object authorization, validation, rate limiting, secrets management, backups and audit logging.
- Data isolation: no direct ERP / SAP / AD integration; only manually entered operational data is stored.

15.2 Authentication and Authorization

- Mobile OTP is cryptographically random, single-use, short-lived, attempt-limited and resend-throttled. Plain OTP is never stored or logged.
- Dashboard credentials use strong password policy and bcrypt / argon2 hashing. Multi-factor authentication is recommended for administrators.
- Server-side RBAC checks role permissions on every endpoint.
- Object-level authorization verifies job assignment, supplier selection, cycle ownership and closure authority.
- Override and final closure endpoints require current-state checks so a stale screen cannot approve an outdated decision.

15.3 Application, Data and Reporting Security

- All traffic uses HTTPS / TLS 1.2+; CORS is restricted to known origins and security headers are enabled.
- Use server-side input validation, parameterised queries / ORM, output encoding and attachment type/size scanning.
- Database and object storage use encryption at rest; report download links are short-lived and access controlled.
- SMS provider, database and object-store credentials remain in a secret store, never in source or client apps.
- Audit login, configuration change, job creation, time selection, supplier response, threshold lock, override, attendance, recirculation, report export and closure.
- Daily encrypted backups and tested recovery procedures are required.

15.4 Threat Model Summary

Threat	Example	Mitigation
Spoofing	Guessing OTP or stealing a session	OTP hardening, short-lived tokens, revocation and device/session monitoring.
Tampering	Changing job/cycle/supplier IDs or attendance	Object-level authorization, state validation, optimistic locking and audit.
Repudiation	Denying override, attendance or closure	Actor/time/IP audit trail and immutable history.
Information Disclosure	Leaking supplier numbers or generated PDF	Least privilege, encrypted storage, short-lived report URLs and minimal SMS content.
Denial of Service / Cost Abuse	OTP or SMS flooding; reminder loops	Rate limiting, resend throttling, queue caps, dead-letter handling and alerting.
Privilege Escalation	Supplier or engineer reaching Procurement override	Server-side RBAC and endpoint-level policy checks.
Race Condition	Third response and cutoff worker run together	Transactional state change, idempotency keys and row/version locking.
Insider Misuse	Unjustified override or config change	Mandatory reason, manager visibility and complete audit log.

15.5 Security Requirements

ID	Area	Requirement	Priority
SEC-01	HTTPS / TLS	All traffic over HTTPS / TLS 1.2+; redirect plain HTTP.	Must
SEC-02	OTP Generation	Secure random OTP; never predictable.	Must

ID	Area	Requirement	Priority
SEC-03	OTP Expiry	Short validity, single-use and attempt lockout.	Must
SEC-04	OTP Throttling	Rate limit requests/resends per number and IP.	Must
SEC-05	OTP Storage	Store only OTP hash; never log plaintext.	Must
SEC-06	Dashboard Passwords	Strong policy; bcrypt / argon2; no plaintext.	Must
SEC-07	RBAC	Enforce all roles on server.	Must
SEC-08	Object Authorization	Verify assignment / ownership for every job, cycle and response.	Must
SEC-09	State Validation	Reject actions invalid for current cycle state.	Must
SEC-10	Idempotency / Locking	Prevent duplicate or racing threshold, cutoff and closure actions.	Must
SEC-11	Input Validation	Validate/sanitise all inputs and uploaded files.	Must
SEC-12	Injection / XSS	Parameterised queries and output encoding.	Must
SEC-13	Secrets	Keep DB/SMS/storage keys in secret store.	Must
SEC-14	Audit	Log security and business-critical actions with actor/time/IP.	Must
SEC-15	Encryption at Rest	Encrypt database, backups and object storage.	Should
SEC-16	Report Access	Authorise exports; use expiring download URLs.	Must
SEC-17	Backup / Recovery	Encrypted daily backups and tested recovery.	Should
SEC-18	Data Retention	Purge OTP and notification logs according to approved retention.	Should
SEC-19	Dependency Security	Patch dependencies and monitor vulnerabilities.	Should

16. Non-Functional Requirements

ID	Area	Requirement
NFR-01	Availability	Target at least 99% availability during operational hours; scheduled workers must resume after restart.
NFR-02	Performance	Dashboard lists and mobile job details should load within 3 seconds under normal conditions.
NFR-03	Timeliness	Threshold confirmation and cutoff decisions should complete within 1 minute of trigger.
NFR-04	Reliability	Duplicate supplier responses, queue retries and SMS callbacks shall not create duplicate confirmations.
NFR-05	Scalability	Support growth in users, suppliers, jobs, cycles, notifications and reports without redesign.
NFR-06	Usability	Field actions use clear labels, minimal typing, confirmation summaries and accessible touch targets.
NFR-07	Localization	Content is template-based to support English, Sinhala and Tamil.
NFR-08	Auditability	Every material decision is traceable across job and cycle history.
NFR-09	Maintainability	Business rules and reminders are configuration-driven and versioned.
NFR-10	Observability	Monitor queue delays, failed SMS, cutoff errors, reminder loops and report failures.
NFR-11	Backup	Daily backup with documented recovery objectives and periodic restore tests.
NFR-12	Privacy	Collect minimum operational PII and avoid exposing unrelated job details in suggestions or SMS.
NFR-13	Accessibility	Readable contrast, scalable text, keyboard-accessible dashboard and screen-reader labels where practical.
NFR-14	Compatibility	Mobile app supports approved Android/iOS versions; dashboard supports current enterprise browsers.

17. Acceptance Criteria

ID	Acceptance Criteria
AC-01	Given Procurement enters a valid PR number and description, when the job is created, then the PR details are stored without creating or changing the ERP PR.
AC-02	Given active engineers exist, when Procurement types part of an engineer name, then matching active engineers appear with department / designation.
AC-03	Given category and sub-category filters are selected, when the supplier list loads, then only active matching suppliers are displayed.
AC-04	Given a filtered supplier result, when Select All is used, then every displayed supplier is selected and can still be removed individually.
AC-05	Given a filtered supplier result, when Procurement searches a name, then the search applies only within that filtered result.
AC-06	Given a job is created, when the assigned engineer opens the app, then the engineer sees only that assigned job and an action to select one time.
AC-07	Given minimum lead time and a three-day window, when the engineer opens the scheduler, then times before the lead time and after the window are disabled.
AC-08	Given a selected supplier has another confirmed visit in the window, when the scheduler loads, then a ranked suggestion is shown without exposing the other PR details.
AC-09	Given the engineer selects one valid time, when it is submitted, then all selected suppliers receive the same time and response request.
AC-10	Given a supplier opens an invitation, when Available is selected, then the response and timestamp are stored for that supplier and cycle.
AC-11	Given two suppliers are already Available, when the third supplier selects Available, then the cycle immediately becomes Confirmed and the time is locked.
AC-12	Given a time is locked, when a later supplier selects Not available, then the selected time remains unchanged and the new response is recorded.
AC-13	Given exactly two suppliers are Available at the cutoff, when the cutoff worker runs, then Override Required appears for Procurement.
AC-14	Given Override Required, when Procurement chooses Proceed without a justification, then the system rejects the action.
AC-15	Given Override Required, when Procurement chooses Proceed with a justification, then the time is confirmed by override and the decision is audited.
AC-16	Given zero or one supplier is Available at cutoff, when the cutoff worker runs, then Reschedule Required is set and the engineer receives an SMS.
AC-17	Given the visit time has passed and the grace period expires, when the engineer has not closed the cycle, then an SMS is sent and repeats at the configured frequency.
AC-18	Given the confirmed supplier list, when the engineer uses Select All and submits attendance, then every supplier receives the selected attendance state.
AC-19	Given attendance is submitted, when the engineer closes the cycle, then Procurement sees attendance and a closure deadline is calculated.
AC-20	Given the Procurement closure deadline expires without action, when the escalation worker runs, then the Procurement Manager receives an SMS.
AC-21	Given Procurement selects Recirculate, when the eligible supplier list opens, then suppliers marked Attended in any earlier cycle of that job are unavailable for selection.
AC-22	Given a prior supplier was Not Available, No Response or Did Not Attend, when the job is recirculated, then that supplier may appear if still active and matching filters.
AC-23	Given recirculation is confirmed, when the new cycle is created, then the parent PR remains the same and the cycle number increases.
AC-24	Given a monthly report is generated, when the PDF opens, then it shows PR, job, cycle, selected time, supplier responses, confirmations, attendance and final outcome.
AC-25	Given an unauthorised supplier or engineer changes a job ID in a request, when the API checks authorization, then access is denied.
AC-26	Given cutoff and third-response events occur at nearly the same time, when both are processed, then only one valid final state and one confirmation notification set are created.

18. Assumptions, Risks and Open Points

18.1 Assumptions

- Procurement receives the PR details before creating the Go See job.
- One active engineer is assigned to each job cycle.
- All engineer and supplier contacts have valid mobile numbers and can receive SMS.
- The normal business expectation is at least three supplier confirmations; selecting fewer than three suppliers requires an agreed exception policy.
- A supplier marked Attended is treated as having participated in the quotation opportunity and is excluded from later cycles for the same job.
- Quotation amount, evaluation and supplier award remain outside Go See.
- The same engineer remains assigned during recirculation unless Procurement explicitly changes the engineer with audit.

18.2 Risks and Mitigations

Risk	Impact	Mitigation
Supplier ignores invitation	Threshold not reached; job delays	Response reminders, dashboard aging and clear cutoff.
Too few suppliers selected	Automatic threshold impossible	Warn below configurable minimum and require exception justification or prevent creation.
Engineer selects time too close	No fair response period	Minimum lead time not shorter than cutoff; urgent exception requires separate policy.
Third response races cutoff	Conflicting decisions	Transaction, idempotency and state locking.
Supplier confirms but does not attend	Operational disruption	Engineer attendance record, no-show reporting and recirculation eligibility rule.
Engineer does not close visit	Procurement cannot complete process	Repeated SMS reminders and dashboard escalation.
Procurement does not close	Open jobs accumulate	Three-day SLA, manager SMS and overdue queue.
Wrong attendance selection	Supplier incorrectly excluded	Engineer confirmation summary; authorised correction with reason and audit.
Too many suggestions	Timeline becomes confusing	Rank confirmed before pending, consolidate duplicates and cap suggestions.
SMS delivery failure	User misses action	Log delivery status, retry policy, app inbox remains source of truth.
Large monthly PDF	Slow generation / unreadable report	Pagination, summary + detailed appendix, asynchronous export and date filters.
Uncontrolled configuration	Business rules become inconsistent	Role restriction, validation, effective dates and audit.

18.3 Open Points for Final Sign-Off

- Confirm whether the active window means 72 rolling hours or the current day plus the next two calendar days.
- Confirm official site working days, working hours, slot increment and minimum visit duration.
- Confirm whether minimum lead time is exactly 24 hours for all jobs or whether urgent jobs need an approved exception.
- Confirm whether Procurement may create a job with fewer than three selected suppliers and what approval / justification is required.
- Confirm the exact supplier response reminder timings before cutoff.
- Confirm whether Procurement can correct engineer attendance after closure and which role approves the correction.
- Confirm whether "Attended" alone is sufficient to treat the supplier as having given a quotation opportunity, or whether a separate Quotation Received field is required.
- Confirm the Procurement Manager recipient(s) and repeat frequency for overdue closure alerts.
- Confirm first-release language coverage for Engineer app and Procurement dashboard.
- Confirm OTP validity, resend cooldown, maximum attempts and retention periods.

Recommended sign-off decision

Approve the process logic first, then confirm the configurable values in Section 12. The architecture allows these values to change without rewriting the process or database model.